

A Low-Power High-Precision Comparator With Time-Domain Bulk-Tuned Offset Cancellation

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Abstract—A novel time-domain bulk-tuned offset cancellation technique is applied to a low-power high-precision dynamic comparator to reduce its input-referred offset with minimal additional power consumption and delay. The design has been fabricated in a commercially available 0.5- μm process. Measurement results of 10 circuits show a reduction of offset standard deviation from 5.415 mV to 50.57 μV , improved by a factor of 107.1. The offset cancellation scheme does not introduce observable offset or noise, and can achieve fast and robust convergence with a wide range of common mode input. Operating at a supply of 5 V and clock frequency of 200 kHz, the comparator together with the OC circuitry consumes 4.65 μW of power, or 23 pJ of energy per comparison.

Index Terms—Analog-digital conversion, bulk-tuned, CMOS analog integrated circuits, comparators, noise, offset cancellation, time-domain.

I. INTRODUCTION

AS THE decision-making circuit that interfaces between analog and digital signals, the comparator is a crucial building block in a wide variety of systems, such as analog-to-digital converters (ADCs) [1]–[5], memory sense amplifiers (SAs) [6], and comparator-based switched-capacitor circuits (CBSCs) [7].

A dynamic comparator utilizes positive feedback to achieve low power, high speed, high gain, and full-swing output. With the current gated by the clock signals, it only consumes dynamic power during evaluation. A key performance metric of a dynamic comparator is its input-referred offset voltage. In many configurations, the offset of the comparator directly affects the performance or yield of a system. Compared to its static counterpart, the dynamic comparator often exhibits a larger offset. The reason is that in addition to static mismatch such as threshold (V_{th}) and trans-conductance (β) mismatch, it also suffers from dynamic mismatch due to imbalance of parasitic capacitors at internal nodes during evaluation [8], [9]. And with transistor size scaled down, these random mismatches impact the offset performance more severely and cannot be relieved by layout strategies. According to [9], in a typical 0.18- μm CMOS latch, a capacitive imbalance of only 1 fF can lead to offsets of several tens of millivolts. A detailed mathematical derivation of mismatch sensitivity of a latched comparator using the perturbation method can be found in [10].

A widely applied approach to reduce the offset is to use linear pre-amplifiers and cancel the offset by negative feedback in auto-zeroing phase [5]. This scheme, however, requires high bandwidth amplifiers with considerable gain, which will inevitably lead to large power consumption. And its effectiveness will degrade with the reduction of drain resistance in deep-sub-micrometer processes. Recent publications have proposed other ways to cancel the offset of comparators.

A common approach is to intentionally introduce imbalances to the circuit to counterbalance the offset. This imbalance can be realized by charge injection at latch nodes [11], binary-weighted capacitor arrays as the load [12], or current sources in parallel with the input pair [13], [14]. Another approach is to tune the threshold of the input pair by using floating-gate input transistors [15], [16], or by trimming the body voltage [17]. In both the counterbalance and threshold-tuning approaches above, the offset is not directly or indirectly measured. Therefore external circuits have to determine the amount of correction either by linear searching or statistically equalizing the output codes of the comparator. The drawback comes from the inherent granularity of the offset correction in these approaches. If the step size of each correction is set small to have a fine resolution, the adaptation will take longer time with large initial offset.

The dynamic switching approach used in [18] uses multi-phase clock to temporarily store the offset in a capacitor, and then subtract it from the input. Based on the same approach, [19] improves the power and offset by using latch load in the first stage. The generation of multi-phase non-overlapping clocks will add to power consumption. And the offset of the second stage is not cancelled in this approach. In [3], the latch can be reconfigured into an amplifier in the auto-zeroing phase thus implementing negative feedback; however it consumes static power in auto-zeroing phase and requires a complex timing control scheme and numerous switches. A time-domain comparator is presented in [2]. By cascading multiple voltage-controlled delay lines, the design trades delay with input-referred noise and offset.

In this paper, we propose a low-power high-precision comparator and an offset cancellation (OC) scheme, first described in [20]. The OC scheme is able to sense the offset in the time domain and eliminate it in closed loop by tuning the body voltages of the input transistors. It can achieve arbitrarily fine resolution and exponential convergence of the residual offset, so that the trade-off between resolution, convergence speed and initial offset range is avoided. In addition, the OC circuitry only requires a single phase clock to operate and adds negligible power and delay to the comparator.

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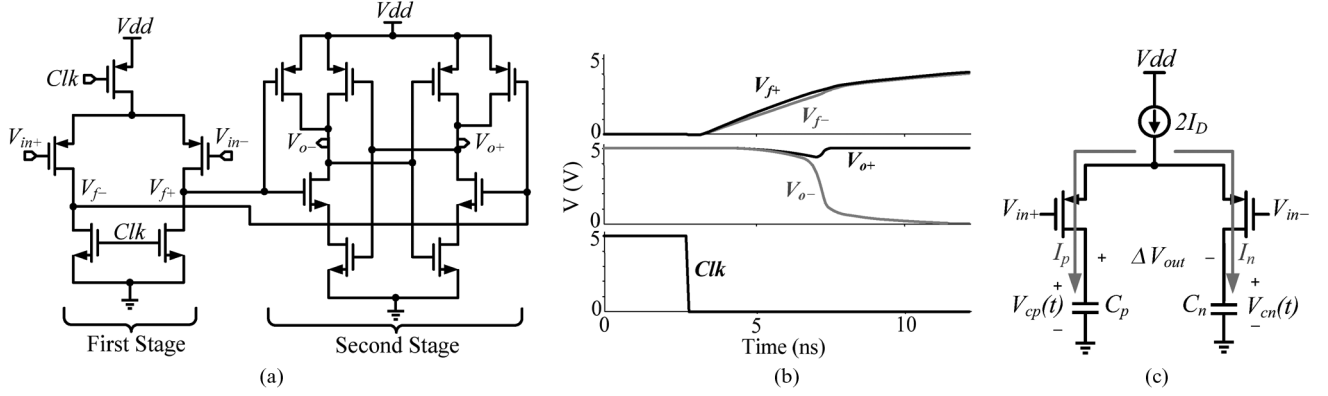


Fig. 1. Two-stage dynamic comparator core. (a) Schematic. (b) Simulation waveforms showing the operation of the comparator with a 10 mV differential input ($V_{in+} > V_{in-}$). (c) Simplified schematic of the first stage during evaluation.

The paper is organized as follows. Section II briefly describes the operation of the two-stage comparator, and how its first gain stage can be designed to minimize the offset and noise of the latch stage. In Section III, the proposed OC scheme is presented. We first describe its architecture and operation principles, and then discuss its advantages and various design considerations. Section IV derives the system model seen by the offset voltage. And based on this model, the criterion for convergence and the offset and noise introduced by the OC circuitry are analyzed. Section V describes the implementation of the circuit blocks in the system. Section VI presents the measurement results of the prototype design and discusses its scaling potential. And Section VII summarizes this work.

II. TWO-STAGE DYNAMIC COMPARATOR

The two-stage dynamic comparator core used in this design, depicted in Fig. 1(a), is inspired by the double-tail voltage sense amplifier [6]. Based on it, several changes are adopted.

Firstly, the second stage is asynchronously clocked by the outputs of the first stage V_{f+} and V_{f-} . This eliminates the need for a complementary clock, and improves resolution by input dependent positive feedback. As a trade-off, propagation delay is increased because V_{f+} and V_{f-} have slower edges compared to a digital clock signal.

Secondly, the input transistors are changed from N to P type. It is found that PFETs have a better flicker noise performance in this process, and the accessibility of the isolated body terminals makes the OC scheme possible. In a triple well process, NFETs can also be used as the input devices.

A. Operation

Before the OC scheme is presented, we briefly describe the operation of the core comparator as background. The comparator comprises two stages. The first stage is a dynamic differential pair to provide gain for pre-amplification of the input signals V_{in+} and V_{in-} . The outputs of the first stage, V_{f+} , V_{f-} function both as reset clock and driving signals for the second stage, which is basically a latch. When clock signal Clk is high, V_{f+} , V_{f-} are reset to low and V_{o+} , V_{o-} high. A falling edge of Clk stops the reset phase and turns on the tail transistor of the first stage; both V_{f+} and V_{f-} rise but at different slopes due to the differential input. When the common mode voltage of V_{f+} and V_{f-} reaches the threshold of the

NFETs in the second stage, the amplification of the second stage takes over and V_{o+} , V_{o-} start to fall; finally the latch activates when the common mode voltage of $V_{o+,-}$ reaches the threshold of the cross-coupled inverters, the positive feedback regenerates the output to a rail-to-rail signal. This process is illustrated by Fig. 1(b).

B. Gain Stage Design

In practice, the accuracy of offset cancellation is limited by the resolution and noise of the comparator core. A substantial gain from the first stage can increase the comparator's sensitivity and attenuate the input-referred noise and offset of the second latch stage. During evaluation, the dynamic voltage gain of the first stage can be derived from the simplified schematic shown in Fig. 1(c).

Assume that during evaluation, the tail transistor can be modeled as a constant current $2I_D$, the channel length modulation effect can be neglected, and the load capacitances on nodes V_{f+} and V_{f-} can be lumped in to C_n and C_p , $C_n = C_p = C$. The input voltages are V_{in+} and V_{in-} , with their difference ΔV_{in} much smaller than the common mode voltage V_{in-cm} .

At the beginning of evaluation $t = 0$, the voltages on the capacitors are $V_{cn}(0) = V_{cp}(0) = 0$, and both the input transistors are in saturation. $V_{cn,p}(t)$ at given time t can be expressed by the differential equations

$$\begin{aligned} V_{cp}(t) &= \frac{1}{C} \int_0^t I_p dt \\ V_{cn}(t) &= \frac{1}{C} \int_0^t I_n dt. \end{aligned} \quad (1)$$

With the assumption that ΔV_{in} is small and that both input transistors are in saturation, the charging currents I_p and I_n can also be expressed as the differential current ΔI and common mode current I_{cm} . $\Delta I = -g_m \Delta V_{in}$, and $I_{cm} \approx I_D$, where g_m is the small signal trans-conductance of the input transistor. With this same assumption, the differential output $\Delta V_{out}(t)$ and common mode output $V_{out-cm}(t)$ can be obtained from (1)

$$\Delta V_{out}(t) = V_{cp}(t) - V_{cn}(t) = -\frac{1}{C} g_m \Delta V_{in} t, \quad (2)$$

$$\begin{aligned} V_{out-cm}(t) &= \frac{1}{C} I_D t, \\ \text{and } \Delta V_{out}(t) &\ll V_{out-cm}(t). \end{aligned} \quad (3)$$

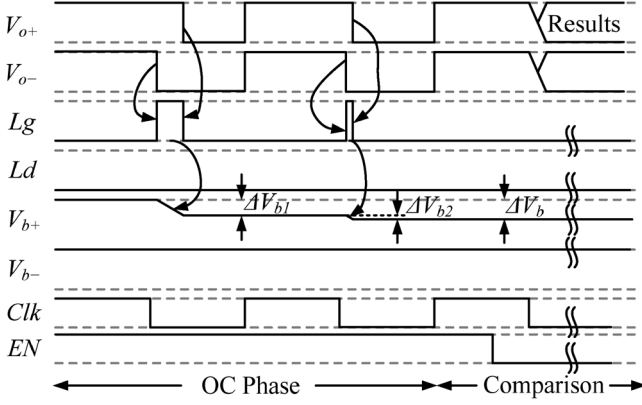


Fig. 4. Typical waveforms during offset cancellation. 2 iterations are shown.

Fig. 4 shows typical waveforms of OC operation. In this case, due to the offset, the falling edge of V_{o+} lags V_{o-} . The PD senses this delay and generates pulses on L_g pin. These pulses turn on one branch of the CP and as a result, the body voltage of one input transistor (V_{b+}) is decreased by ΔV_b while the other (V_{b-}) remains constant. By the same token, if V_{o+} leads V_{o-} due to opposite polarity of offset, V_{b-} will be decreased in a similar manner with V_{b+} unchanged.

C. Resolution and Convergence Speed

During OC, the comparator is operating in a way very similar to normal operation. Only a minimum of components are added to the comparator core to enable self-calibration. Moreover, the offset is sampled at the output, which encompasses all the sources of mismatch inside the comparator. The offset introduced by the OC circuitry is negligibly small in most cases and will be discussed in Section IV. Therefore, the proposed OC scheme can achieve accurate cancellation of the offset.

The OC scheme forms an iterative loop. Each time it is executed, it generates an amount of correction proportional to the magnitude of measured offset. Therefore, it can have arbitrarily fine resolution of offset correction while the approaches in many previous works [12]–[17] have inherent granularity and therefore limited resolution determined by the step size of each correction.

The OC action can be repeated multiple times (two times in Fig. 4). Each time the offset will be reduced to a fraction of its previous value. The offset decreases exponentially with the number of iterations. Compared to [12]–[17] in which the offset decreases linearly, the proposed scheme can achieve faster convergence speed even with large initial offset.

D. Common Mode Input Range and Retention Time

The change of body voltage needed to cancel the offset mainly depends on the magnitude of initial offset and γ in (6). Its worst case value can be found from Monte-Carlo simulation. In the simulation, OC is executed to calibrate out the comparator offset due to mismatch, and the body voltage change at the end of OC phase is recorded for each run. In the 200 Monte-Carlo runs we performed, the largest body voltage change is 67 mV, which will not cause body diode to turn on. And the input common mode range will not be affected.

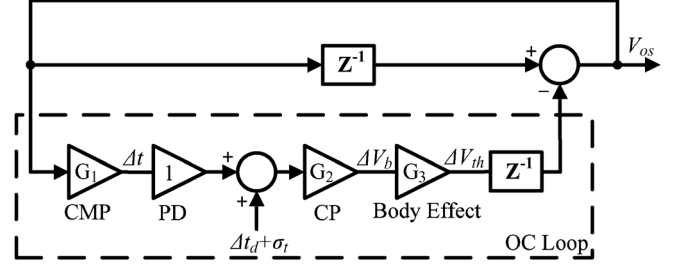


Fig. 5. The system model seen by the input-referred offset V_{os} . The upper loop models the retention of the offset in the storage capacitors. The OC circuitry is modeled as a cascade of gain stages and a unit delay.

Since the body voltages are stored on capacitors, the retention time is simulated to determine the refreshing rate required to maintain a good offset performance. At 55 °C temperature, the rate of change of V_{b+} and V_{b-} is about -112 mV/s. It should be noted that this voltage droop is common mode to both V_{b+} and V_{b-} , and the difference of them which decides the OC accuracy changes more than a magnitude slower at a rate of -9.5 mV/s after cancelling a 10 mV offset. Therefore, a very low refreshing rate (<100 Hz) is acceptable. The OC circuitry does not draw static current and only needs to be enabled at a very small duty cycle, so additional power consumption from OC circuitry is negligible. On the other hand, thanks to its fast convergence and low power consumption, the proposed OC scheme can be repeated frequently without consuming excessive time or power. In this way, not only offset, but also slow-varying time-correlated noise such as flicker noise can be reduced just as in auto-zeroing amplifiers [22].

IV. SYSTEM MODELING AND FEEDBACK LOOP DESIGN

During OC, a discrete-time feedback loop is formed in order to null the offset voltage. To ensure a robust design, the convergence criterion of the loop needs to be determined. This can be done by modeling the system seen by the input-referred offset voltage V_{os} as in Fig. 5.

In Fig. 5, the upper loop models the retention of the offset value in the storage capacitors. The OC loop enclosed by the dashed line consists of four gain stages. The unit delay is added because the change of body voltage happens after the output of the comparator settles and therefore is effective at the next clock cycle. The first gain stage G_1 models the comparator's conversion from differential input V_{os} to the delay Δt between its outputs during the OC phase. The PD's gain can be considered to be unity. However it introduces delay mismatch Δt_d and timing jitter σ_t as will be discussed later. The CP converts pulse width output from the PD to the change of the body voltage ΔV_b by the ratio of $G_2 = I_T/C_{b+,-}$, where I_T is the tail current of the CP. And by using first order approximation of (6), the gain due to body effect can be expressed as $G_3 = |\partial V_{TH}/\partial V_b| = \gamma/(2\sqrt{2}\phi_F)$.

A. Convergence Criterion of the OC Loop

First ignoring the error terms Δt_d and σ_t for simplicity, the difference equation associated with V_{os} at given time step n can be expressed as

$$V_{os}(n) = V_{os}(n-1) - G \cdot V_{os}(n-1), \quad (7)$$

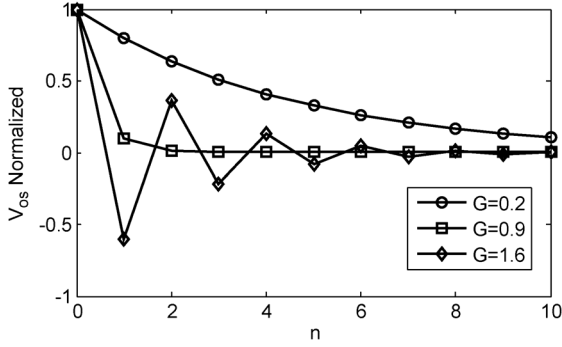


Fig. 6. The evolution of $V_{os}(n)$ with different G values.

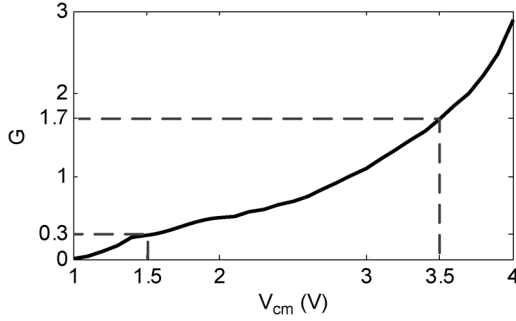


Fig. 7. The loop gain G at different input common mode voltage V_{cm} .

where G is the product of all the gains in the OC loop, $G = G_1 G_2 G_3$. Solving (7) with initial condition of $V_{os}(0)$ gives

$$V_{os}(n) = (1 - G)^n V_{os}(0). \quad (8)$$

The convergence of $V_{os}(n)$ requires that the base of the exponent term to be less than 1 in magnitude. Therefore we get the convergence criterion of the OC scheme

$$0 < G < 2. \quad (9)$$

The evolution of $V_{os}(n)$ with different G values are plotted in Fig. 6 ($V_{os}(n)$ is normalized with respect to $V_{os}(0)$). As can be seen from the plot, a G close to 1 produces a fast convergence of V_{os} towards 0, while $G > 1$ leads to overshoot. The overshoot is undesirable because it will cause more reduction of body voltages than needed. In general, the number of iterations needed to achieve 1% residual offset is

$$n_{0.01} = \text{ceil}(\log_{|1-G|} 0.01), \quad (10)$$

where $\text{ceil}(x)$ is the nearest integer greater than or equal to x .

Among the three components of G , G_3 is fixed given a certain process; G_2 can be tuned by changing the storage capacitors size or the CP tail current; and G_1 is inherent to the comparator design. G_1 increases with the common mode input V_{cm} . Because at higher V_{cm} , the overdrive for the input transistors is lower, so they stay longer in saturation region during evaluation and provide more amplification. For the same reason, V_{cm} also affects the dynamic voltage gain, as suggested by (5). To achieve reliable convergence over a wide range of V_{cm} , the loop gain G is designed to be 0.85 when $V_{cm} = 2.5$ V. The resulting loop gain at different V_{cm} extracted from post-layout simulation is plotted in Fig. 7. From Fig. 7, with the V_{cm} varying from 1.5 V to 3.5 V, G is in the range of 0.3 to 1.7, which ensure a reduction of 99%

offset in less than 13 iterations. When V_{cm} is in the range from 2.8 to 3.0, only 2 iterations are needed to reduce offset to 1%.

B. Offset and Noise Introduced by the OC Circuitry

In simulation, it is found that the offset and noise introduced by the OC circuitry is dominated by the delay mismatch Δt_d and timing jitter σ_t in the Δt signal path, especially after a few iterations when Δt is already reduced to a small value. Taking them into consideration, (7) can be rewritten as

$$V_{os}(n) = (1 - G)V_{os}(n-1) - G_2 G_3 (\Delta t_d + \sigma_t). \quad (11)$$

Solving it yields

$$V_{os}(n) = (1 - G)^n V_{os}(0) - G_2 G_3 \left(\Delta t_d \frac{1 - (1 - G)^n}{G} + \sigma_{t-tot} \right), \quad (12)$$

where the first term is the residual offset after n -th iteration, the second term is the error caused by Δt_d and σ_t , σ_{t-tot} is the contributions of stochastic timing jitters of each iteration summed in power. At the end of OC phase, $(1 - G)^n$ converges to 0, therefore the error term can be simplified with its final value

$$V_{os_err} = -G_2 G_3 \left(\Delta t_d \frac{1}{G} + \sigma'_{t-tot} \right), \quad (13)$$

and the standard deviation of σ'_{t-tot} is $\sqrt{1/(1 - (1 - G)^2)} \sigma_t$.

An important observation from (13) is that in order to minimize the effect of mismatch and jitter of the OC circuitry, the gain of CP (G_2) need to be low. As a result, the gain of comparator (G_1) need to be high to keep G unchanged. Increasing G close to 2 seems to improve the offset because of the first term in the parenthesis. However, this will lead to severe overshoot as discussed above, and also increase the noise σ'_{t-tot} . So a G close to 1 is desirable for both convergence speed and accuracy.

C. PVT Variations of Loop Gain G

As discussed above, the deviation from the point of $G = 1$ can affect accuracy and convergence speed. As shown in Fig. 7, $G = 1$ when $V_{cm} \approx 2.8$ V. Therefore, the optimum performance can be obtained at common mode input of 2.8 V. In most applications, V_{cm} is a fixed value, and the designer can tune either G_1 or G_2 to achieve $G = 1$ at the common mode input in interest. To show the robustness of the circuit with PVT variations, Monte-Carlo simulations were performed at combinations of temperature variation of 0–55 °C and supply voltage variation of $\pm 10\%$. The G value is extracted by comparing the output delay at the first and second iteration in OC phase. In these simulations, V_{cm} is 2.8 V for nominal supply voltage of 5 V and varies in proportion to the supply, which is realistic in most supply variation cases. The resultant mean and standard deviation values of G are summarized in Table I. The worst-case mean variation of G occurs at $V_{dd} = 4.5$ V and 0 °C. With additional 3σ of process variation, the worst-case G over PVT variations is determined to be around 1.25, which is 25% deviation from its nominal value of 1. In this worst case, the number of iterations needed to achieve 1% residual offset is 4; the overshoot is not significant; and the effect of mismatch and jitter of the OC circuitry is still kept low according to (13).

TABLE I
 PVT VARIATIONS OF LOOP GAIN G

Supply Voltage (V)	Temperature (°C)					
	0		25		55	
	μ	$\sigma \times 100$	μ	$\sigma \times 100$	μ	$\sigma \times 100$
4.5	1.214	1.238	1.145	1.192	1.071	0.970
5	1.137	1.006	1.085	0.938	1.027	0.844
5.5	1.043	0.771	1.006	0.730	0.963	0.718

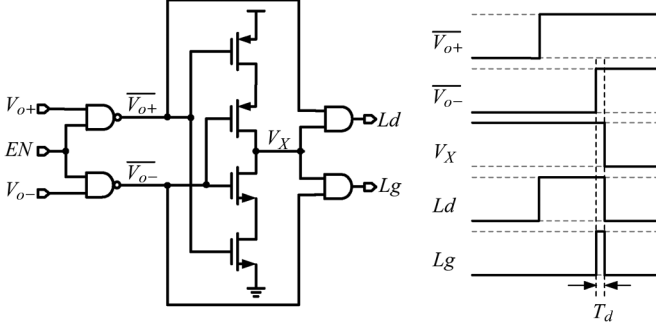


Fig. 8. Schematic and timing diagram of the PD. Storage V_X will not change its state unless both $\overline{V}_{o+}$ and $\overline{V}_{o-}$ change states. If the tri-state inverter has a small propagation delay T_d , pulses will be generated on both Ld and Lg outputs, therefore eliminating the dead zone.

V. CIRCUIT IMPLEMENTATION

A. Phase Detector

The function of the PD is to sense the delay between its two inputs. Conventional PDs based on latches or flip-flops as in [23] are not adequate for application in this design, because most of them suffer from dead-zone, which will contribute to residual offset. Moreover, since they operate in closed loop, the speed is limited and the output jitter due to metastability [23] will add to random input-referred offset.

A simple yet novel dynamic PD circuit is proposed to meet the requirements of the OC scheme. The schematic and timing diagram is shown in Fig. 8. The PD is enabled by the EN signal and the outputs from the comparator are buffered with two NAND gates. The inverted signals $\overline{V}_{o+}$ and $\overline{V}_{o-}$ are then sent to a structure similar to a tri-state inverter. Note that the storage node V_X will not change its state unless both $\overline{V}_{o+}$ and $\overline{V}_{o-}$ change states. Therefore, V_X is the inversion of the one of the inputs that comes later. The two AND gates will then generate pulses corresponding to the polarity and magnitude of delay. Especially, if the tri-state inverter has a small propagation delay T_d , pulses are always generated on both Ld and Lg outputs, and the difference of the two pulse widths equals the magnitude of input delay. This effectively eliminates the dead zone.

The simple structure of the proposed PD makes it very power and area efficient. Open loop operation improves the speed and jitter performance. Post-layout simulation shows that the proposed PD is able to distinguish a 2.8 ps timing difference, which translates to less than 5 μ V input-referred offset of the comparator.

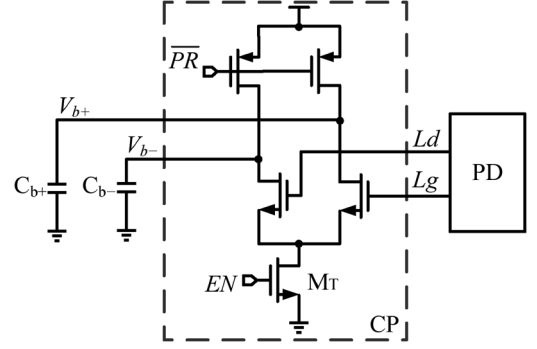


Fig. 9. Schematic of the charge pump.

B. Charge Pump and Storage Capacitors

The charge pump is implemented as in Fig. 9, shown together with the PD and the storage capacitors holding the body voltages of comparator input pair (C_{b+} and C_{b-}).

Before OC starts, C_{b+} and C_{b-} are precharged to V_{dd} by a PR pulse. Then the CP is enabled by the EN signal, the two transistors connected to the PD will discharge either C_{b+} or C_{b-} with pulse width controlled by the offset. The discharging current is set by the tail transistor M_T , a differential pair structure ensures the matching between the two current branches.

The size of the storage capacitors is 10 pF each, which is limited by two factors. During operation, the charge on the capacitors will slowly leak through the junctions of the input transistors as well as the CP transistors [24], and this determines the achievable retention time. However, the effect of this leakage is relatively small because most of it is common mode to the input pair. The other factor is the kickback noise from the input pair. During evaluation, the voltage swing on the drain and source will be coupled to the storage capacitor by the diffusion-to-body parasitic capacitors C_{db}/C_{sb} . This kickback is mostly common-mode to the comparator when a small differential input is to be resolved, and will not affect the results given the symmetry of the comparator. The differential components due to the mismatch of $C_{b+, -}$, and C_{db}/C_{sb} will be captured by the OC circuitry and reduced together with the offset. This is shown in the Monte-Carlo simulation result of residual offset after OC in Fig. 10. This simulation includes the process variation and mismatch of all the components in the design. To further show that the circuit is insensitive to the unbalanced kickback, we intentionally introduced 10% mismatch to the $C_{b+, -}$, as well as 20% mismatch to C_{db} . With these extreme (and unlikely) mismatches added, the same Monte-Carlo simulation was run again and the circuit is still able to maintain a low residual offset. The results are summarized in the table in Fig. 10.

VI. MEASUREMENT RESULTS

The proposed design was fabricated in a 0.5- μ m CMOS process. The die photo is presented in Fig. 11. Two identical circuits were instantiated on the same chip. The active area is 0.064 mm², more than half of which is occupied by the 10 pF storage capacitors C_{b+} and C_{b-} realized with double-poly. In more recent processes in which metal-insulator-metal (MIM) capacitor of higher density is available, one can stack the

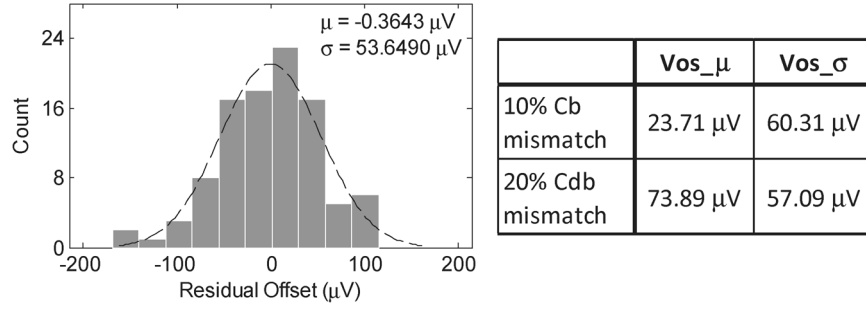


Fig. 10. Monte-Carlo simulation results of residual offset. The results with extra mismatch sources are listed in the table.

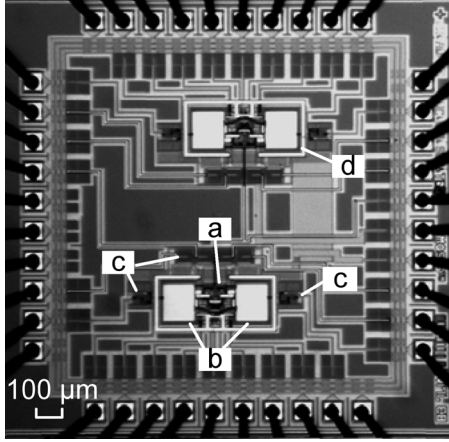


Fig. 11. Micrograph of the comparator with OC circuitry, fabricated in a 0.5- μm process. (a) Comparator core with OC circuitry except storage capacitors. (b) Storage capacitors. (c) Output buffers. (d) Another identical instantiation of the circuit.

storage capacitors above the active area and significantly reduce the area overhead.

Unless otherwise specified, all the following measurements are made with 5 V supply, clock frequency of 200 kHz, common mode input V_{cm} of 2.8 V, four clock cycles of OC phase and at room temperature.

During normal operation, the comparator (not including the output buffers) consumes 0.93 μA current from the 5 V supply. In OC phase, the current consumption is 0.96 μA , indicating that the OC circuitry consumes negligible power. Due to the fact that there is no static current, the power consumption of the whole system scales down linearly with clock frequency until it is dominated by the leakage. The energy per comparison is 23 pJ/comp.

The propagation delay from clock to output is measured to be 15 ns, corresponding to a 33.3 MHz maximum operation frequency. The comparator is current starved to improve resolution as described in Section II. A higher speed can be obtained by increasing the gain stage tail current at the cost of resolution.

A. Operation and Convergence of the OC Scheme

The operation waveforms of the comparator and the OC circuitry are shown in Fig. 12. The outputs of the comparator are buffered by geometrically scaled inverter chains, and the body voltages by unity gain buffers before connected off-chip. Fig. 12(a) shows 4 clock cycles' OC operation followed by normal comparisons. The comparator outputs V_{o+} , V_{o-} at the

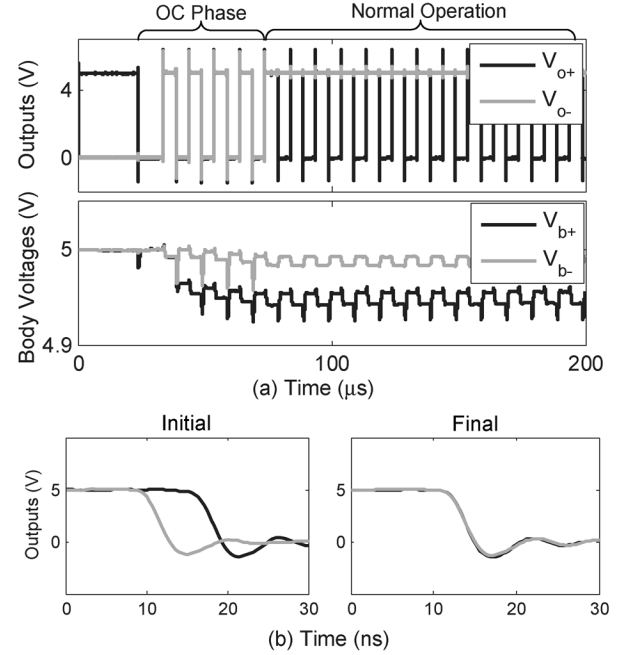


Fig. 12. (a) Oscilloscope waveforms of the comparator outputs V_{o+} , V_{o-} , and the body voltages of the input transistor V_{b+} , V_{b-} . The voltage steps in V_b are due to the kickback, as discussed in Section V-B. (b) Details of V_{o+} , V_{o-} before the OC is executed (initial), and after 3 iterations of OC (final). The ringing in the waveforms are artifacts due to the parasitics of the measurement setup.

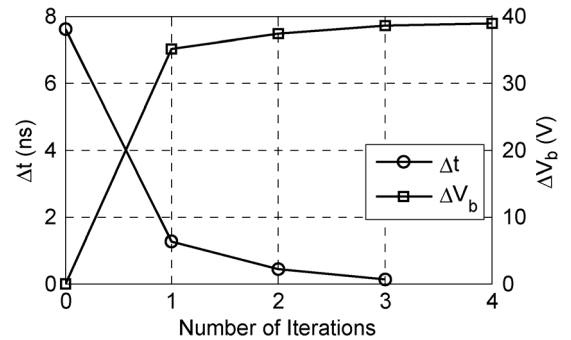


Fig. 13. The evolution of the output delay Δt (delay between V_{o+} and V_{o-}), and body voltage difference $\Delta V_b = V_{b-} - V_{b+}$ with the number of OC iterations.

first and last falling edge of clock during OC phase is shown in details in Fig. 12(b). Initially at the first edge, V_{o+} lags V_{o-} by about 7 ns due to the comparator offset. After 3 iterations of OC, the two outputs overlap with each other, indicating a very low residual offset. The evolution of the output delay Δt (delay between V_{o+} and V_{o-}), and body voltage difference $\Delta V_b = V_{b-} - V_{b+}$ during OC iteration is plotted in Fig. 13,

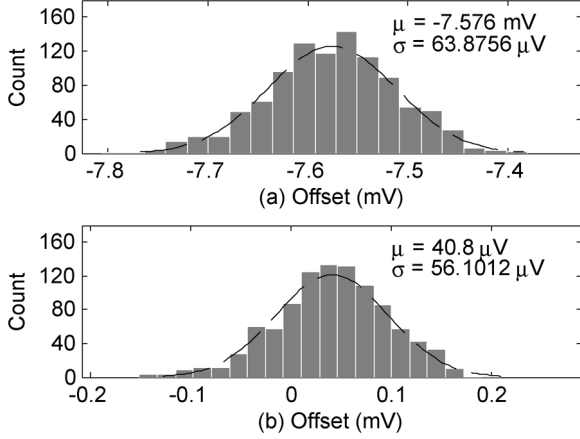


Fig. 14. The histograms of trip points (a) without the OC scheme enabled, (b) with the OC scheme enabled.

showing that the OC scheme is able to converge to its final value exponentially within 4 clock cycles. Due to the one-clock-cycle delay between Δt and OC action, the Δt at the 4th iteration was not measured because that would require a 5th clock cycle in OC phase.

B. Offset Distribution Among the Chips

The test setup to characterize the offset of the prototype chips is built around a National Instrument data acquisition (DAQ) card. The DAQ card is programmed to provide the control signals (Clk , EN , PR) necessary for the comparator and OC circuitry to operate. To measure the offset, one of the comparator inputs is supplied with a reference voltage, while the other is connected to a ramp signal generated by an off-chip integrator. The ramp starts from below the reference voltage and increases with time. Both inputs are also connected to the analog input channel of the DAQ card in differential mode. The output of the comparator is wired as the sampling trigger of the DAQ card so that the differential input when the comparator output toggles (the trip point) is sampled. This measurement is repeated 1000 times, and we interpret the mean value of these 1000 trip points as the measured offset, while the variance of them are attributable to the noises from the comparator core, OC circuitry and the measurement system. The offset and noise from the measurement system, which are smaller than the comparator's offset and noise, are obtained by doing the same measurement with the DAQ card inputs shorted. This procedure is done each time before a measurement is made to account for the drift of measurement system. In all the following results, the measurement system offset and noise have been subtracted from the raw data.

The trip point histograms of a comparator without and with the proposed OC scheme enabled are plotted in Fig. 14. It can be seen that the OC scheme does not degrade the noise performance, while reducing the offset from -7.58 mV to 40.8 μ V. In fact, the noise performance is improved because the OC scheme also cancels out time-correlated flicker noise [22].

The offset distribution of 10 comparators on 5 chips was measured with and without the OC scheme enabled. The statistical results are listed in Table II. The offset standard deviation

TABLE II
OFFSET STATISTICS WITHOUT AND WITH OC AMONG 10 CIRCUITS

	μ	σ	Min	Max
Without OC	396.2 μ V	5.415 mV	-8.325 mV	5.587 mV
With OC	28.30 μ V	50.57 μ V	-40.25 μ V	99.68 μ V

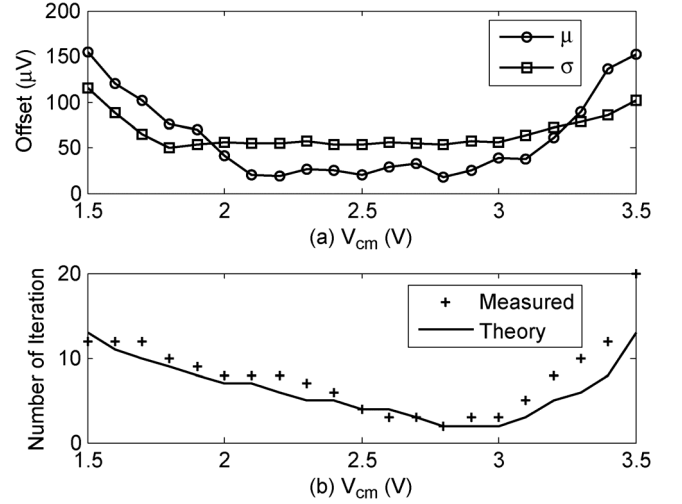


Fig. 15. OC performance at different common mode input V_{cm} . (a) The input-referred offset and noise of the comparator with OC enabled. (b) The number of iterations to achieve a 1% residual offset.

among the 10 circuits is reduced by a factor of 107.1, corresponding to a 40.59 dB improvement in SNR or a 6.74 bits improvement in resolution. The residual offset after OC is dominated by the offset and mismatch introduced by the OC circuitry, and is comparable to the input-referred noise level of the comparator, indicating a very effective offset cancellation.

C. OC Performance at Different V_{cm}

As discussed in Section IV, the performance of the proposed OC scheme is dependent on the common mode input V_{cm} , because the loop gain G varies with V_{cm} . The OC performance at different V_{cm} is measured and plotted in Fig. 15.

Fig. 15(a) shows the measured offset and input-referred noise of the comparator with OC at different V_{cm} after convergence. As predicted by (13), both the offset and noise introduced by the OC circuitry increases as the loop gain G deviates from 1 due to V_{cm} change. However, the degradation in performance is not significant until the deviation is large. From Fig. 15(a), both the input-referred offset and noise is smaller than 100 μ V as long as V_{cm} is in the range of 1.8 V to 3.3 V.

According to (10), the convergence speed decreases when G deviates from 1, and Fig. 15(b) shows this effect. The solid line is the number of iterations needed to achieve a 1% residual offset predicted by (10) and Fig. 7, and the plus signs are the measured values. The measurement and theory match each other closely, which validates the analysis in Section IV. When V_{cm} is in the range of 2.5 V to 3 V, the proposed OC scheme is able to converge to 1% residual offset in 4 clock cycles or less. The optimum V_{cm} can be tuned to accommodate different applications, as discussed in Section IV.

TABLE III
PERFORMANCE SUMMARY AND COMPARISON

	This work	TCSI'05 [15]	TCSI'08 [16]	ASSCC'09 [18]	ASSCC'11 [13]	CICC'08 [11] ³
Technology	0.5 μm	0.35 μm	0.35 μm	90 nm	90 nm	65 nm
Active Area	64k μm^2	2509 μm^2	13.7k μm^2	152 μm^2	1330 μm^2	1625 μm^2
Energy consumption	23 pJ/comp.	2.7 pJ/comp.	2.75 pJ/comp.	78 fJ/comp. ¹	61 fJ/comp.	380 fJ/comp.
Maximum clock frequency	33.3 MHz	100 MHz	1.2 GHz	500 MHz	1.5 GHz	1 GHz
Energy consumption normalized	34.09	20.77	21.15	35.94 ¹	28.11	242.04
Max clock frequency normalized	32.68	53.94	648.65	82.57	247.73	119.27
Input-referred offset σ	50.57 μV	379 μV	199 μV	3.8 mV	553 μV	-
Input-referred noise	56.10 μV_{rms}	742 μV_{rms}	314 μV_{rms}	-	320 μV_{rms}	-
Offset Convergence Speed	Exponential	Linear	Linear	One Cycle ²	Linear	-
Refreshing Rate	100Hz	Non-volatile	Non-volatile	Each Cycle	Not Stated	External Control
Use Dedicated Calibration Cycle	Yes	Yes	Yes	No	Yes	-

¹ Simulated energy consumption.

² Dynamic switching, offset is cancelled for each cycle.

³ Offset programming only, OC not included.

D. Performance Summary and the Effect of Scaling

Table III summarizes the performances of the prototype comparator with offset cancellation circuitry and compares them to several recently reported works.

It is worth noting that both the proposed comparator and OC scheme are very amenable to process scaling because of their dynamic and time-domain operation. While the accuracy of the circuit in [5] will be degraded due to the reduced drain resistance, the proposed scheme will actually benefit from smaller feature size because it can provide smaller parasitic capacitance and higher time resolution [25]. The design is also tolerant to supply scaling because there is no stacking of transistors (such as those in cascode structure). In order to compare performance of designs implemented in different technologies, we include normalized speed and power results. In Table III we normalized the maximum clock frequency by the delay of a single inverter at the same technology node, and the energy per comparison by the switching energy of that same inverter. The process parameters were extracted from the foundry test results representative of the specific technology node. Given that the dynamic comparator operates essentially similar to digital gates, normalizing its performance with that of an inverter in the same technology node can provide figures of merit independent of the process in which the circuit is implemented. Compared to the previous works in Table III, this work achieves the best offset and noise performance at the cost of lower clock speed. The exponential convergence ensures a fast and robust offset settling. And the normalized energy per comparison is comparable to the state-of-art.

To further investigate the technique's scaling potential, the circuit was implemented and simulated in a 90-nm CMOS process with 1.2 V supply voltage. The Monte-Carlo simulation results show a residual offset standard deviation of 95.3 μV , which is consistent with the results in the 0.5- μm process above, while the offset standard deviation without OC is 6.19 mV, indicating an improvement factor of 65. Due to the shrinking of power supply and parasitic capacitance, the energy consumption is reduced drastically to 47.8 fJ/comp. and the maximum clock frequency is increased to 180 MHz. Noting that all the results are from schematic simulations, we expect these performances (mainly energy consumption and maximum

clock frequency) to degrade in a real circuit. The bulk voltage storage capacitors $C_{b+,-}$ are designed to be 2 pF each. Using Hi-K MIM capacitors in this process, the total area of these two capacitors is smaller than 700 μm^2 . Based on the scaling factor, the area of the active devices is estimated to be slightly less than the capacitors area. And with the storage capacitors stacked over the active devices, the total area of the circuit is same as the capacitor area. Due to increased leakage current in this deep-sub-micrometer process, the minimal refreshing rate required at 55 °C is about 2 kHz, which is still a small fraction compared to the increased clock frequency. A lower refreshing rate can be achieved by using larger storage capacitors at the cost of area.

VII. CONCLUSION

We have presented a low-power high-precision dynamic comparator with a novel time-domain bulk-tuned offset cancellation scheme in a 0.5- μm process. The comparator is optimized to minimize the noise and offset. The proposed OC scheme samples the offset in the time domain and cancels it in a closed-loop fashion with bulk-tuning technique. It consumes negligible power, does not sacrifice speed, or require complex timing control. And it can simultaneously achieve fine resolution and fast convergence even with large initial offset.

Measurement results show that with the proposed OC scheme, the offset standard deviation of 10 comparators is reduced from 5.415 mV by a factor of 107.1 to 50.57 μV . The OC loop is able to converge to 1% of its final value in less than 4 clock cycles within proper input common mode range. The comparator with OC circuitry can operate at 33.3 MHz maximum clock frequency with energy consumption per comparison of 23 pJ, and is well-suited to take advantage of process scaling.

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